

AI-1 Spring 26 Project Proposal

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Idea Short Name (N-Puzzle AI Solver using A* Search)

Idea Description

This project solves the **N-Puzzle** problem (e.g, 8-puzzle) using artificial intelligence.
The system takes an initial scrambled state of tiles and finds the optimal sequence of moves to reach the goal state using the **A* search algorithm** with heuristics like **Manhattan distance**.

Agent Architecture

- ☐ Simple Reflex Agent
- ☐ Model-Based Reflex Agent
- ✓ ☒ Goal-Based Agent
- ☐ Utility-Based Agent

Problem-Solving Framework

- ✓ ☒ Classical Search
- ☐ Local Search (Optimization)
- ☐ Adversarial Search (Two-Player Games)

☐ Constraint Satisfaction Problem

Proposed Algorithm(s)

- **A* Search Algorithm**
- **Manhattan Distance Heuristic**
- **(Optional) Breadth First Search for comparison**
- **Priority Queue (Min-Heap using `heapq`)**

PEAS

Performance Measures	• Find shortest solution path • Minimize number of moves • Fast execution time • Correctness of final state
Environment	• N x N puzzle board (e.g., 3x3) • Fully known initial state • Goal state predefined
Actuators	• Move tile up • Move tile down • Move tile left • Move tile right
Sensors	• Current board state • Position of blank tile(0) • Goal state comparison

Environment Properties

☒ Fully Observable ☐ Partially Observable	☒ Deterministic ☐ Stochastic ☐ Strategic	☒ Discrete ☐ Continuous
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☒ Single-Agent ☐ Multi-Agent	☒ Sequential ☐ Episodic	☒ Static ☐ Dynamic
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